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Battery-Aware Scheduling for Multi-Receiver Laser Wireless Power Transfer in IoT Systems

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Abstract

This work proposes a scheduling scheme for a single-transmitter (Tx) multi-receiver (Rx) laser-based optical wireless power transfer (OWPT) system. Each Rx represents its battery state using a 3-bit LED combination, which is recognized by the Tx through a camera. Based on the battery-state information obtained in the previous frame, the charging time is adaptively allocated so that Rxs with lower battery states receive more charging time. Experimental results show that proposed scheme mitigates voltage degradation in low-battery Rxs and alleviates energy imbalance among Rxs.

Introduction

- **Wireless Sensor Networks (WSNs)**
 - WSNs rely on **battery-powered** sensor nodes
 - **Battery depletion** limits long-term network operation
- **Optical Wireless Power Transfer (OWPT)**
 - **OWPT** can remotely supply energy to sensor nodes
 - **Laser-based OWPT** enables **selective power transfer** due to its high directionality and narrow beam divergence
- **Need for Multi-Receiver (Rx) Scheduling**
 - If Rxs have different battery states, **low-battery Rxs** should be charged with **higher priority**
 - Therefore, efficient **charging-time allocation** is required to mitigate **energy imbalance** among Rxs

System Model

- **Laser-Based OWPT System Testbed**
 - The system consists of a **single fixed transmitter (Tx)** and **multiple fixed Rxs**
 - The Tx includes a laser power transfer module, a pan-tilt camera, and a control unit
 - Each Rx consists of a PV cell, battery, retroreflective film, and 3-bit LED module

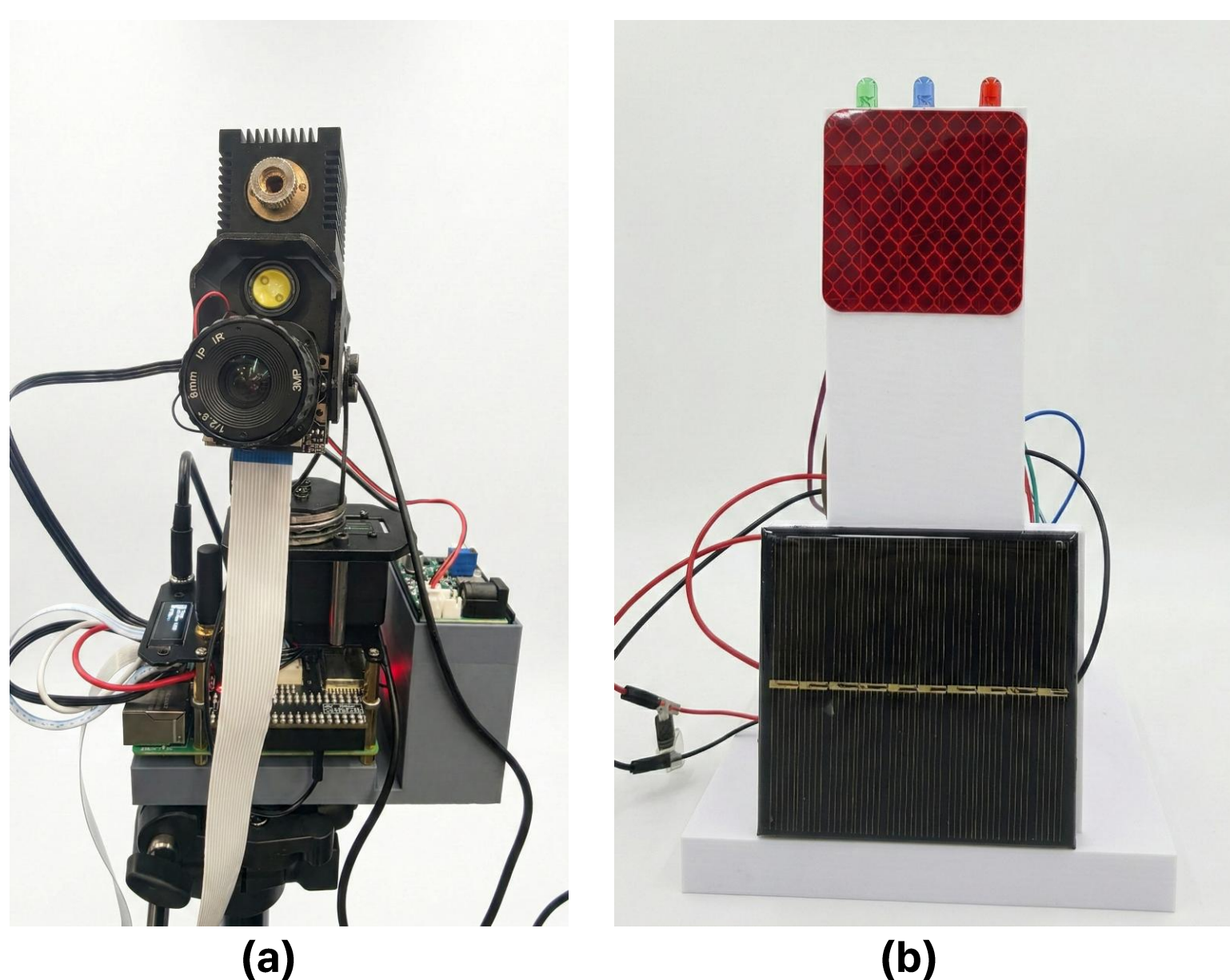


Fig. 1. Photographs of the Tx and Rx modules used in the experimental laser-based OWPT testbed : (a) Tx, (b) Rx.

- **System Operation**
 - The Tx identifies Rx positions and battery states using **camera-based recognition**
 - Each Rx represents its battery state using a **3-bit LED combination**
 - The operation consists of two stages, scanning and scheduling
 - Time-division charging is performed, where **only one Rx is charge at a time**

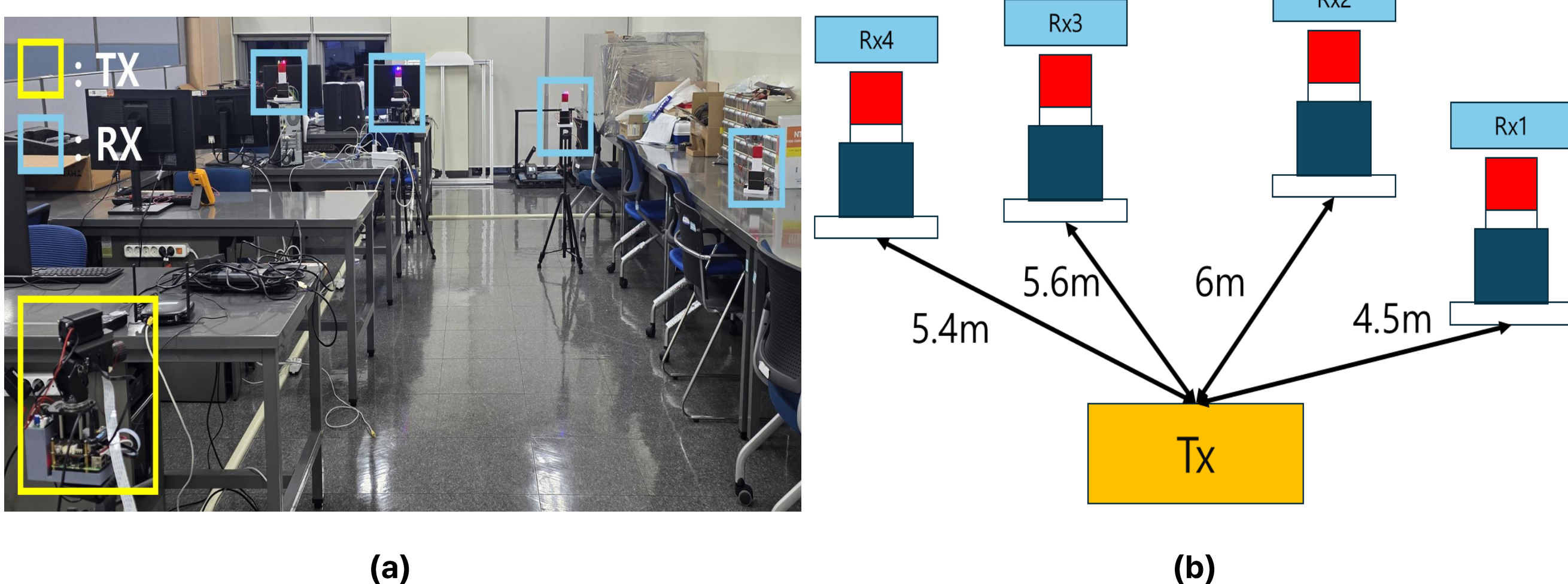


Fig. 2. Experimental laser-based OWPT testbed: (a) Photograph of the indoor test environment, (b) Layout of the experimental laser based OWPT testbed.

Proposed Scheme

- **Proposed Battery-State Based Scheduling**
 - Each Rx represents its **battery state** using a 3-bit LED combination
 - The Tx recognizes the battery state of each Rx through **camera-based recognition**
 - A **lower battery state** produces a larger **charging coefficient**
 - Charging time is allocated according to the coefficient ratio in each frame
 - Therefore, low-battery Rxs receive **more charging time** than high-battery Rxs
- **Conventional Round-Robin Scheduling**
 - Equal charging time is allocated to all Rxs in each frame
 - However, it **does not consider the battery state** of each Rx

Performance Evaluation

- **Experimental Setup**
 - A total of four Rxs are evaluated using a dataset acquired from the **implemented indoor laser-based OWPT testbed**
 - The Tx-Rx distances range from 4.5 m to 6 m
 - The frame duration is set to 240 s, and 8 frames are evaluated
 - The initial 3-bit battery states are set to 100, 101, 110, and 101

Table. 1. System parameters.

Laser wavelength	850 nm
Number of Rxs	4
Tx-Rx distance for each Rx	4.5~6 m
Number of frames	8
Duration of a single frame	240 s

- **Charging-Time Allocation**
 - In the **round-robin scheme**, **60 s** is allocated to each Rx
 - In the **proposed scheme**, charging times of **80, 60, 40 and 60 s** are allocated according to the battery state
 - Rx1 is allocated the longest charging time because it had the lowest initial battery state
- **Result Analysis**
 - In the proposed scheme, Rx1 voltage **increased by approximately 5 mV**
 - In the round-robin scheduling, Rx1 voltage **decreased by approximately 5 mV**
 - The proposed scheme mitigates voltage degradation in the low-battery
 - This result shows that adaptive charging-time allocation can alleviate **energy imbalance** among Rxs

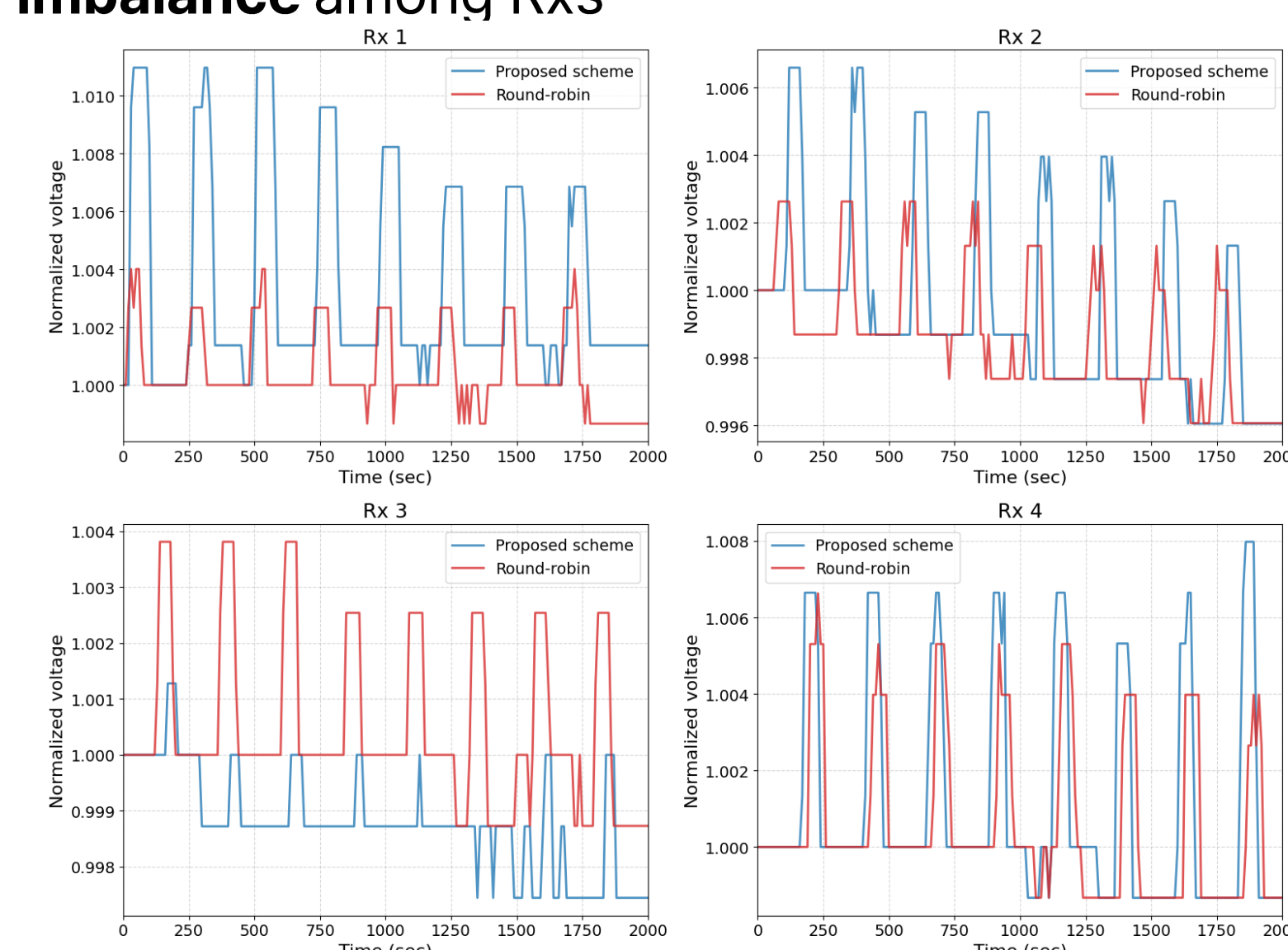


Fig. 3. Comparison of normalized battery-voltage trajectories of the Rxs under the proposed and round-robin scheduling schemes.

Conclusion

- In this paper we propose a **battery-state based scheduling** scheme for multi-Rx laser based OWPT systems
- The proposed scheme allocates more charging time to low-battery Rx
- Experimental results show that the proposed scheme mitigates voltage degradation in low-battery Rx
- Therefore, the proposed scheme can alleviate energy imbalance